

Tornadoes Hitting Hospitals: Data Mining and Preliminary Analysis

Introduction and Background

Tornado danger unrecognized:

- Design for tornadoes not added to International Building Code until 2024. Continued reluctance to implement requirements and design guidelines, due to misperceptions about tornado impacts.

Our role to help:

- Raising awareness by making database of critical facility impacts
 - Critical Facility:** a building that provides essential services to the community (e.g. hospitals, schools)
 - Impact:** the tornado went onto the property or grounds of the facility
- Our team's vision is to eventually make a publicly accessible interactive web map of the database.

Abstract and Motivation

Methods: Data Mining of Hospital Impacts

- Tools: MySQL; Google Sheets, Earth, & Maps

Methods: Research and Classification

- How was the hospital affected?
- Tools: Google Sheets, Search, Earth, & Maps

Weakness to Envelope Damage: We Hypothesize...

- Similar reduced hospital capacity % occurrence between structural and envelope damage
- Envelope damage could have later consequences
- Would suggest a need for greater attention to strength of hospital building envelopes

Results

Fig. 1: Occurrence of Reduced Hospital Capacity, by Damage Type

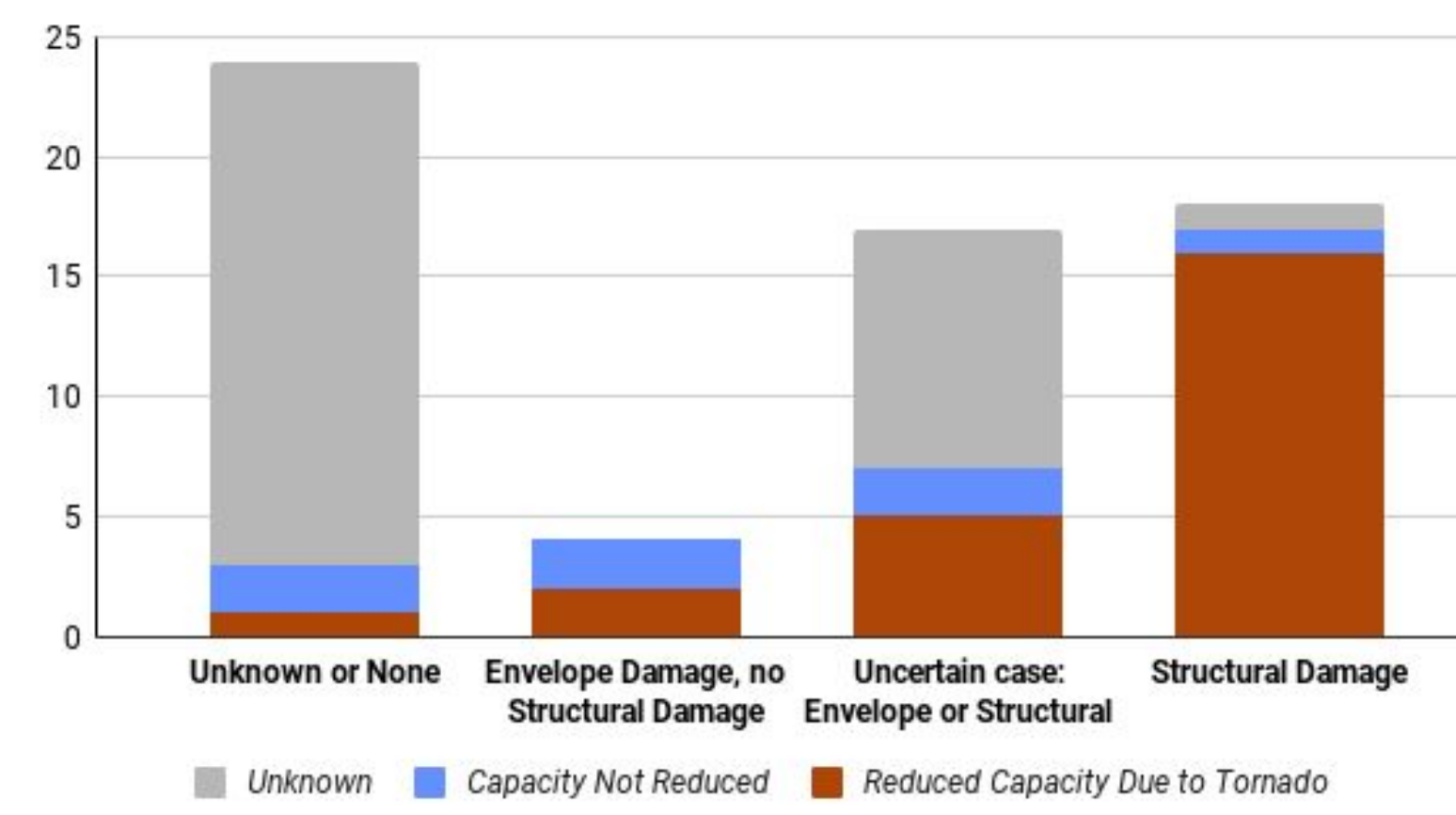


Fig. 2: Occurrence of Reduced Hospital Capacity (Total)

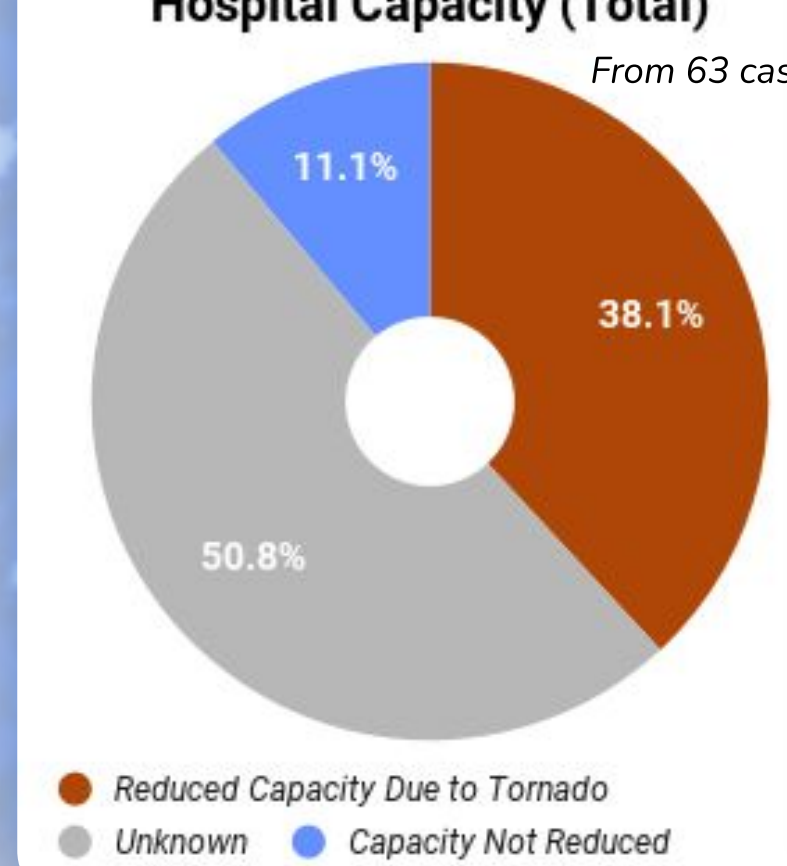


Fig. 3: Occurrence of Reduced Hospital Capacity (Structural Damage)

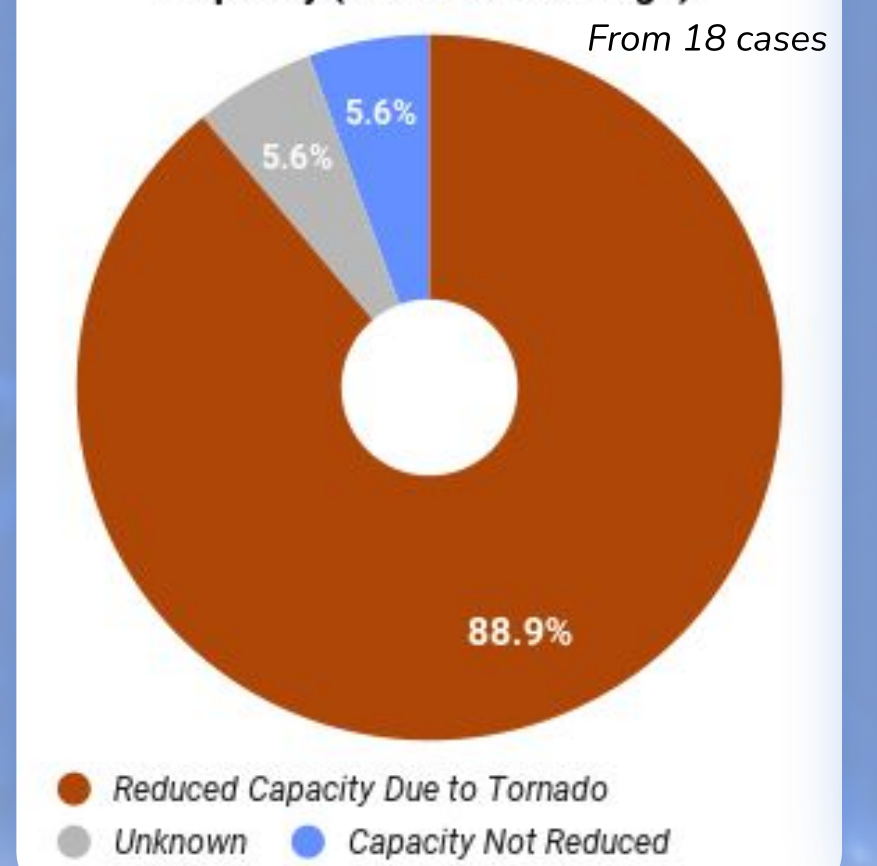


Fig. 4: Occurrence of Reduced Hospital Capacity (Envelope Damage, no Structural Damage)

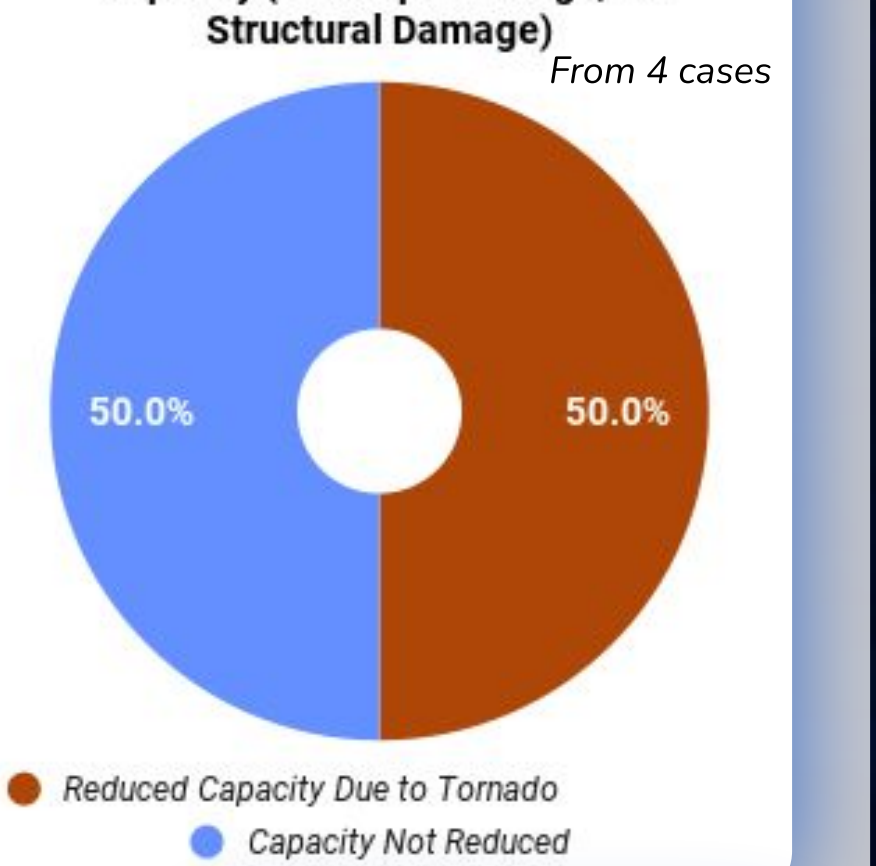


Fig. 5: Hospital Damage Map

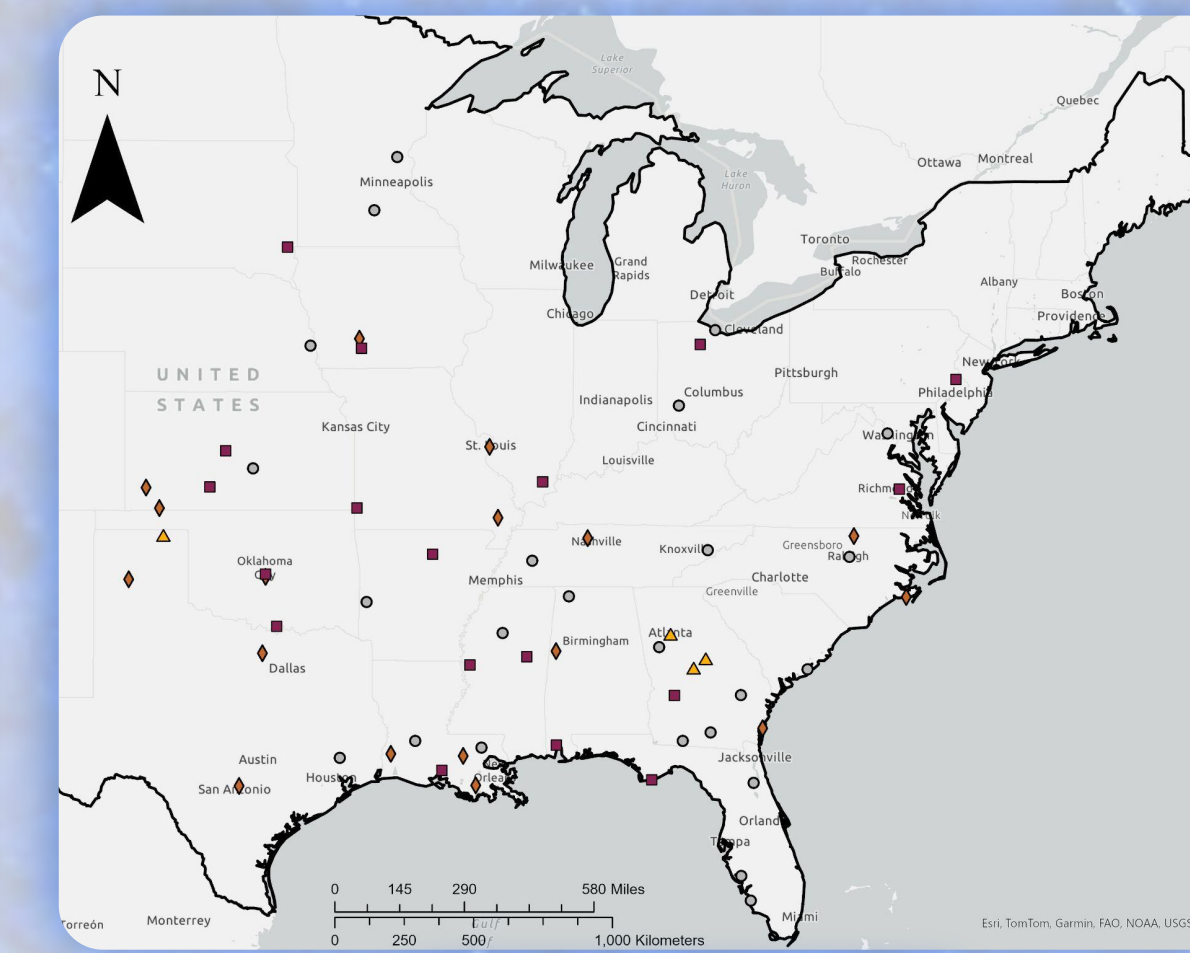


Fig. 6: Hospital Reduced Capacity Map

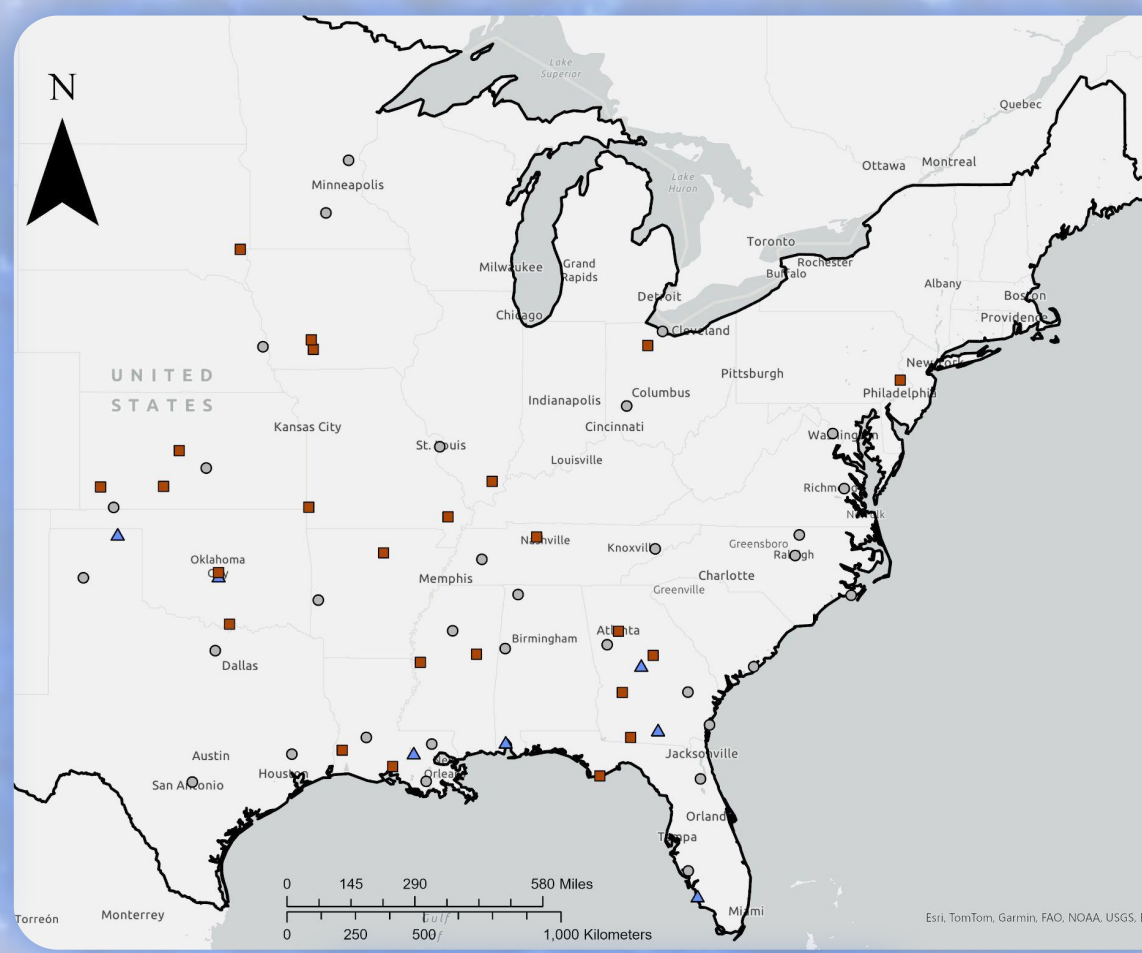


Fig. 7: Occurrence of Envelope and Structural Damage, by Tornado Intensity

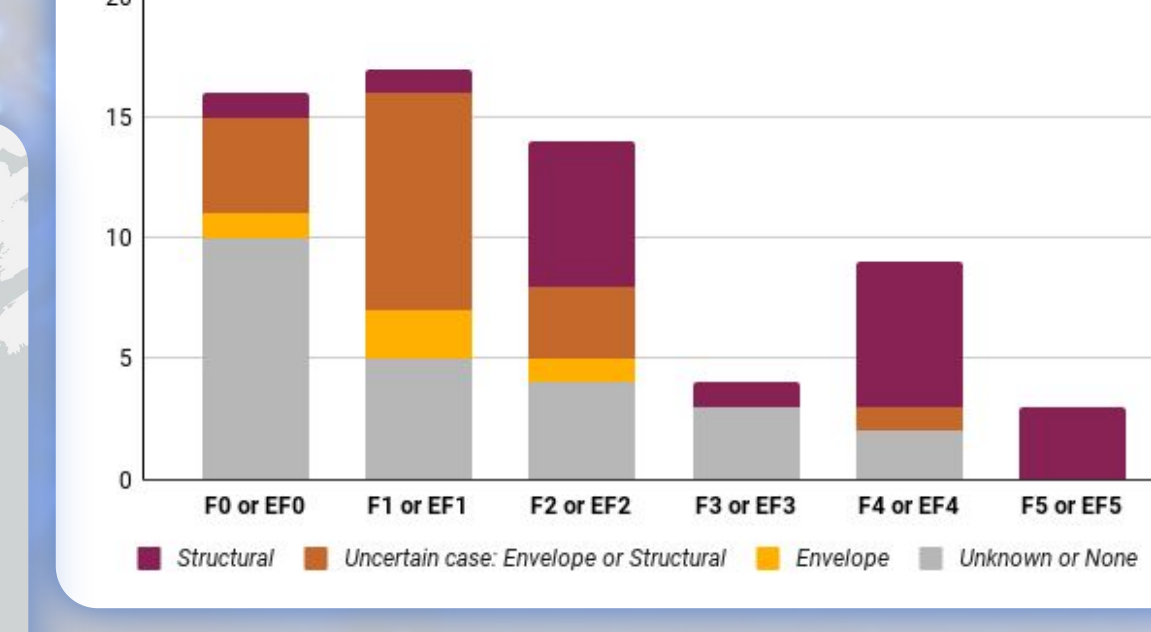


Fig. 9: Occurrence of Damage Types (Reduced Capacity)

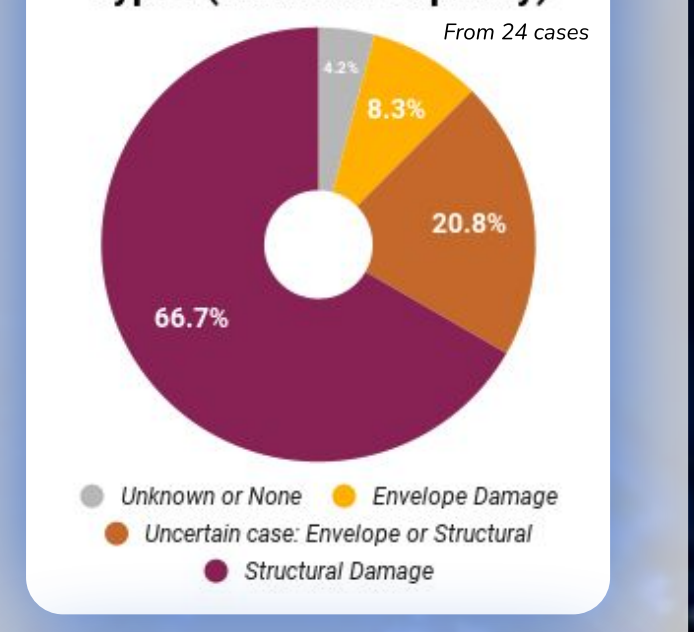


Fig. 8: Occurrence of Capacity Reduction, by Tornado Intensity

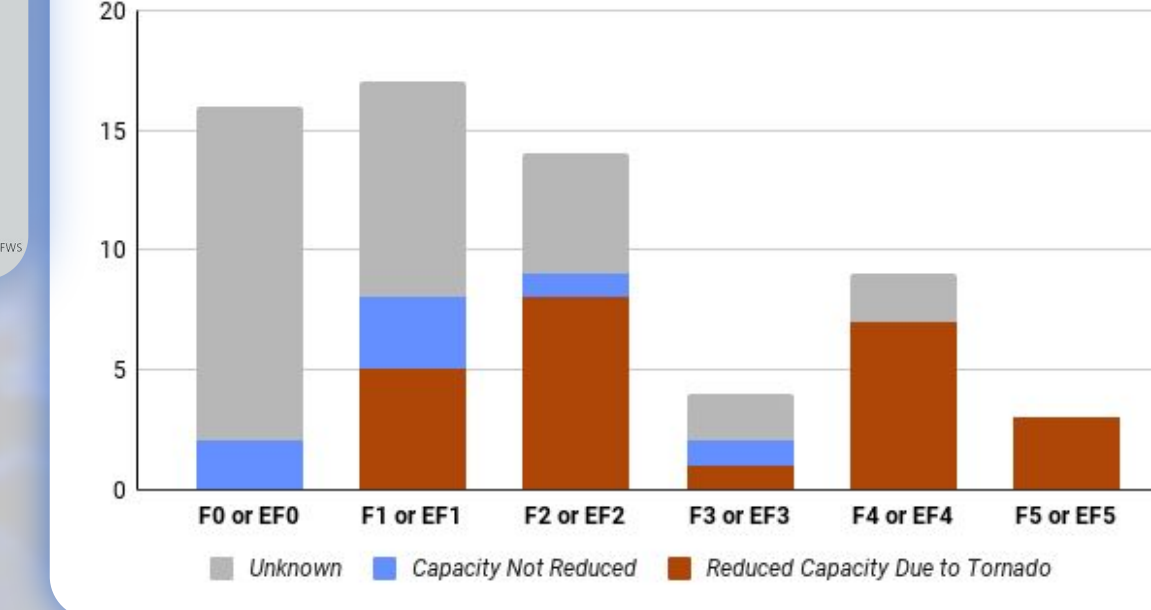
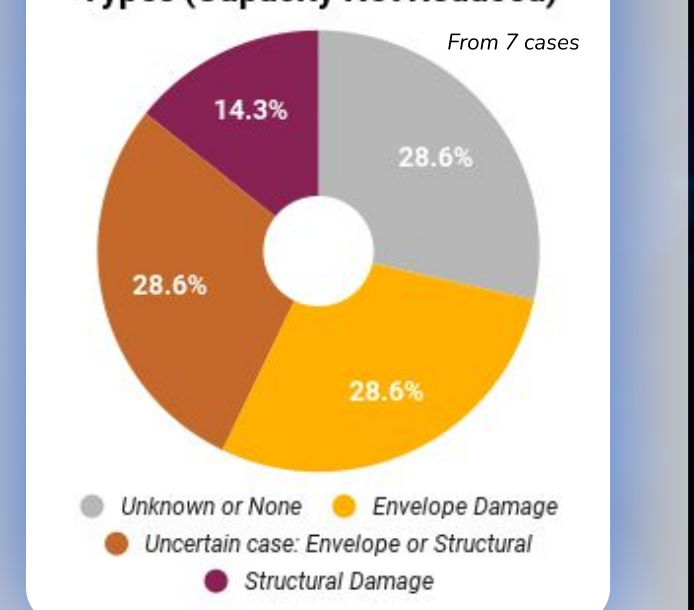
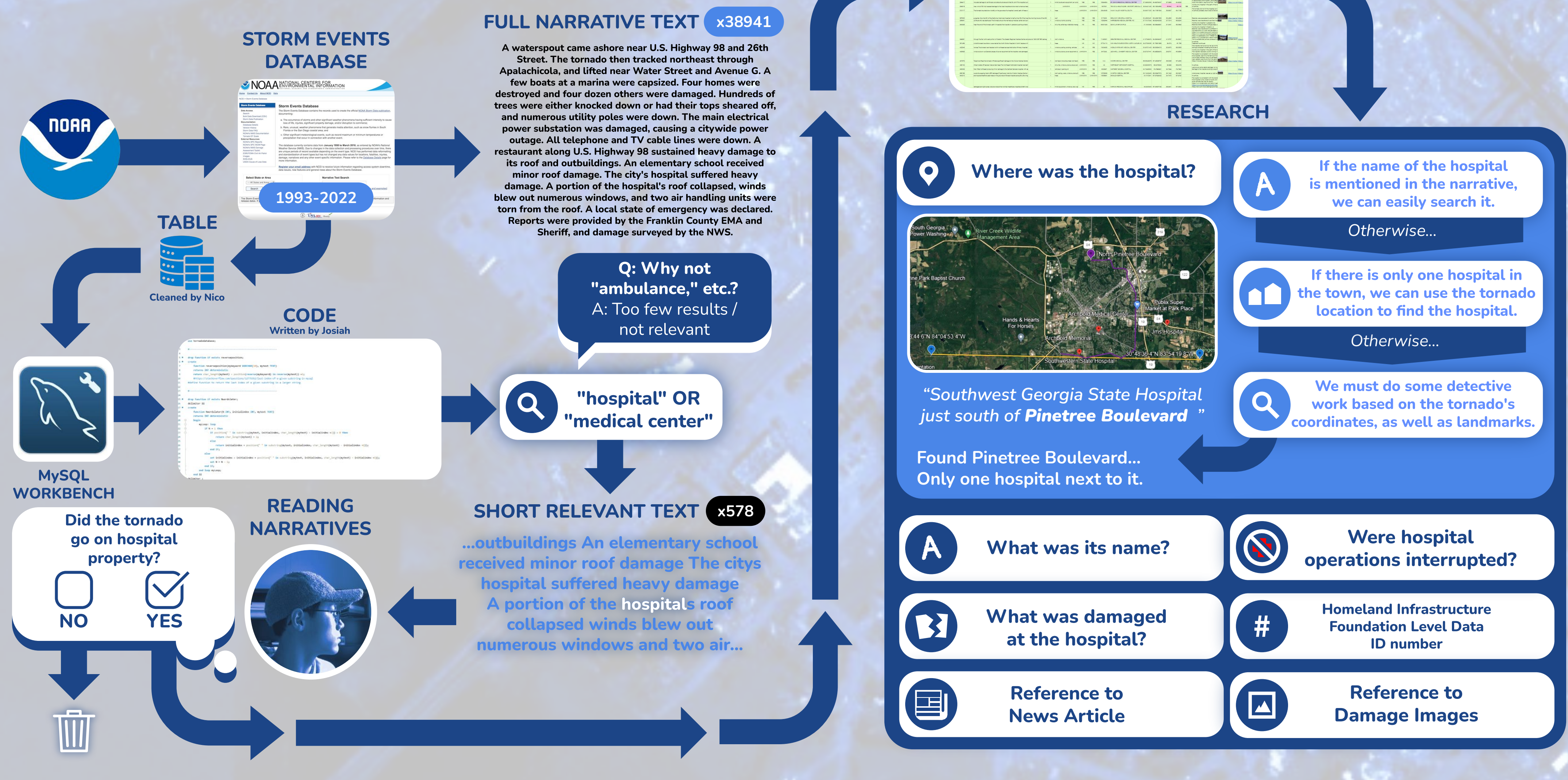


Fig. 10: Occurrence of Damage Types (Capacity Not Reduced)



Data Mining Methods



Research and Classification Methods



Discussion

Danger to Hospitals

- Identified 63 hospital campuses struck by tornadoes 1993-2024: overall average about two per year.
- One in ~107 hospitals have been hit by a tornado.

Resilience to Envelope Damage

- Among four cases of exclusively envelope damage:
 - Why only four? Verified no structural damage.
 - % occurrence of reduced hospital capacity was lower than among cases of structural damage.
 - % occurrence of capacity not reduced was higher than among cases of structural damage.
- This suggests that even envelope damage can incapacitate hospitals: protecting the envelope is crucial. Even so, surprisingly many hospitals continued operations despite envelope damage.

- Most media report hospital status shortly after the tornado. Sometimes repercussions of envelope damage take longer to appear.

How Can Design Guidelines Help?

- New guidelines cover approx. EF0-EF2 wind speeds – more than half of the hospitals which had reduced capacity due to the tornado could have avoided damage by adopting these guidelines.

Uncertainty Analysis

- Sample size 63.
- High number of unknowns: 32 capacity, 23 damage.
- Corresponds to small number of hospitals existing in USA: 6,120-8,013 hospitals vs. over 27,000 fire stations, over 98,000 schools
- Mysterious lack of F3/EF3 tornadoes.
- Reporting bias: surveyors, media report hospital conditions more often when hospital is damaged.

Acknowledgements

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Project Teammates: Nelson Tucker, Silvia Brown, Kevin Ly, Katherine Tang

References

NOAA, National Centers for Environmental Information (NCEI), Storm Events Database (SED), (Years 1993 through Feb 2022), Washington, DC, NOAA, 2022.
<https://www.ncdc.noaa.gov/stormevents/details.jsp>.

Oracle Corporation, MySQL Workbench 8.0, Austin, TX, 2024.